



2010

Preliminary Course

FINAL EXAMINATION

Tuesday 14th September

Mathematics

General Instructions

- o Reading Time - 5 minutes.
- o Working Time – 3 hours.
- o Write using a black or blue pen.
- o Approved calculators may be used.
- o All necessary working should be shown for every question.
- o Begin each question in a new booklet.

Total marks (120)

- o Attempt Questions 1- 10.
- o All questions are of equal value.

Outcomes to be Assessed:

A student:

- P2** provides reasoning to support conclusions which are appropriate to the context.
- P3** performs routine arithmetic and algebraic manipulation involving surds, simple rational expressions and trigonometric identities.
- P4** chooses and applies appropriate arithmetic, algebraic, graphical, trigonometric and geometric techniques.
- P5** understands the concept of a function and the relationship between a function and its graph.

Question 1	(12 Marks)	Use a Separate Booklet	Marks
a)	Expand and simplify: $(x-3)(2x+1)$		2
b)	Simplify $\sqrt{27} + \sqrt{12}$		2
c)	Express the decimal $0.\dot{2}\dot{1}$ as a fraction in simplest form.		2
d)	Factorise the following expression fully: $4x^2 - 16$		2
e)	Rationalise the denominator of: $\frac{\sqrt{3}}{3-\sqrt{2}}$		2
f)	Solve for x : $\frac{2x-1}{5} = \frac{3x+2}{4}$		2

End of Question 1

Question 2 (12 Marks)	Use a Separate Booklet	Marks
a) Is the function $f(x) = 8x^3 - 7x$ even, odd or neither? (Show working).		2
b) For the function $y = 2^x + 1$:		
i) Sketch, showing all the essential features.		2
ii) State the domain and range.		2
c) Sketch the graph of $f(x) = x^2 + 5x + 6$, showing all essential features		3
d) Show the region of the number plane where the following hold simultaneously (do not find points of intersection):		3
$x^2 + y^2 \leq 9$		
$y < x$		

End of Question 2

Question 3 (12 Marks)

Use a Separate Booklet

Marks

- a) A rock climber is standing on the top of a vertical cliff which is 80 metres above sea level. The angle of depression of a tanker out to sea is 32° .

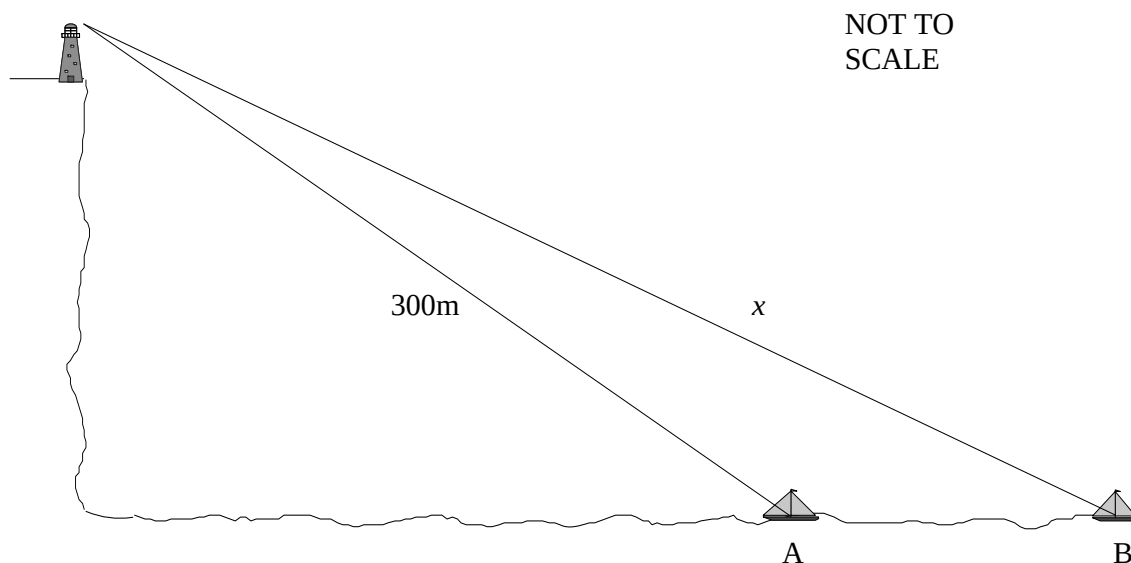
i) Draw a diagram to show this information

1

ii) Calculate the distance (to the nearest metre) from the base of the cliff to the tanker.

2

- b) Two yachts, 250 m apart, are both due east of a lighthouse, which is on top of a cliff. The angle of elevation of the lighthouse from Yacht A is 37° . It is known that Yacht A is exactly 300 m from the lighthouse.



(Assume the cliff meets the water at right angles)

i) Using the cosine rule, show that x is 522 m (to the nearest metre).

2

ii) Using the sine rule find the angle of elevation (to the nearest minute) of the lighthouse from Yacht B.

2

Question 3 (continued)**Marks**

c) Solve $2\cos^2\theta = 1$ for $0^\circ \leq \theta \leq 360^\circ$

3

d) Solve for x : $4^{2x-1} = 8$

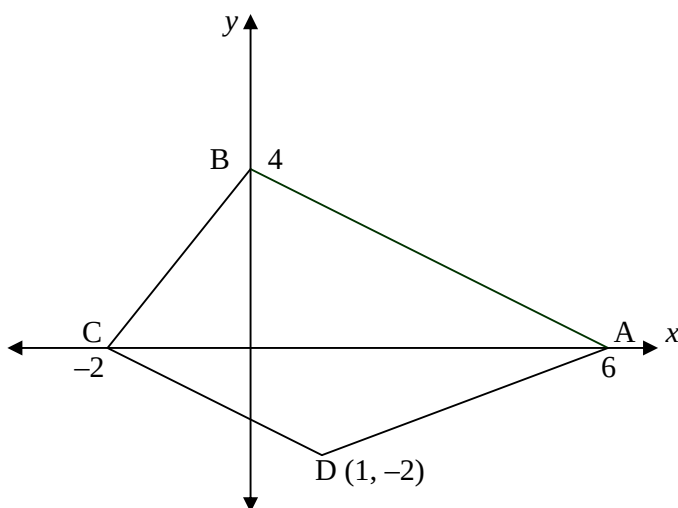
2**End of Question 3**

Question 4 (12 Marks)

Use a Separate Booklet

Marks

a)



ABCD is a Trapezium as shown in the diagram above.
The coordinates of the point D are $(1, -2)$.

- | | | |
|-------|---|----------|
| (i) | What acute angle does the line BC make with the positive x-axis?
Give your answer to the nearest degree. | 2 |
| (ii) | Find the midpoint (M) of the interval AB. | 1 |
| (iii) | Show that the equation of the line CD is given by $2x + 3y + 4 = 0$. | 2 |
| (iv) | Show the distance from the point M to the line CD
is equal to $\frac{16\sqrt{13}}{13}$ units. | 2 |
| (v) | Given that CD is $\sqrt{13}$ units, calculate the distance AB and hence
find the area of the trapezium ABCD. | 3 |

- | | | |
|----|---|----------|
| b) | Sketch $y = \sin x$ for $0^\circ \leq x \leq 360^\circ$ | 2 |
|----|---|----------|

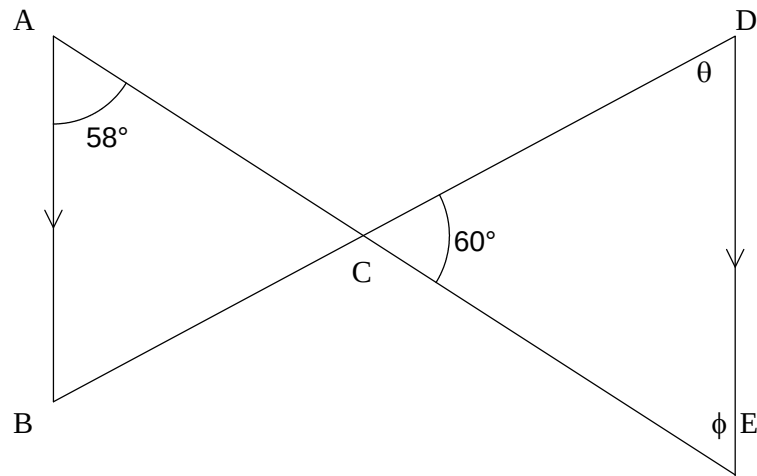
End of Question 4

Question 5 (12 Marks)

Use a Separate Booklet

Marks

- a) Find ϕ and θ , giving reasons.

2

- b) If $f(x) = \begin{cases} -3, & x \geq 0 \\ 2x - 3, & x < 0 \end{cases}$

Evaluate $f(-1) + f(1)$.

2

- c) Factorise $a^3 + 8$

2**Question 5 continues**

Question 5 (continued)**Marks**

d) The sum of the interior angles of a regular polygon is 3960° .

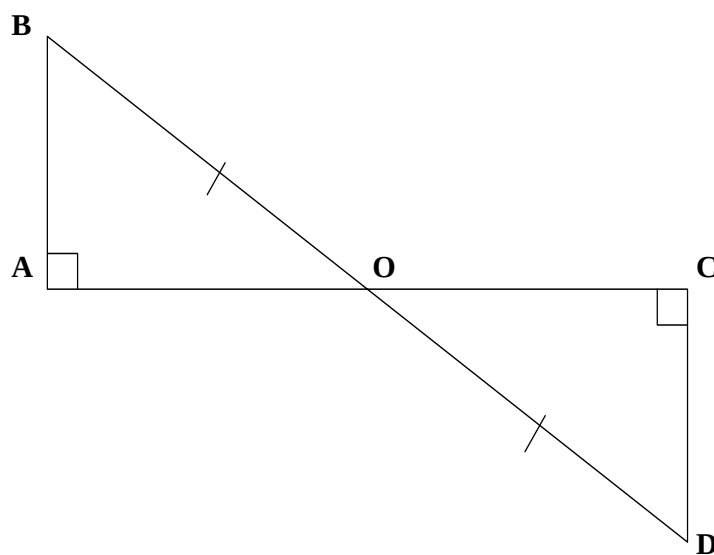
i) Show that the number of sides of the polygon is 24.

1

ii) Find the size of each interior angle.

1

e)



Prove that $\triangle AOB \cong \triangle COD$.

2

f) Solve $|3x - 5| \leq 2$

2**End of Question 5**

Question 6 (12 Marks)	Use a Separate Booklet	Marks
a) Solve the following, leaving answers in exact simplified form: $x^2 - 6x + 7 = 0$		3
b) Solve $(5^x)^2 - 26(5^x) + 25 = 0$ for x .		3
c) Show that $-3x^2 + x - 1 < 0$ for all x .		2
d) For the equation $2x^2 + 5x - 3 = 0$ with roots α and β , find the value of:		
i) $\alpha + \beta$		1
ii) $\alpha\beta$		1
iii) $\frac{1}{\alpha} + \frac{1}{\beta}$		2

End of Question 6

Question 7 (12 Marks) Use a Separate Booklet **Marks**

a) By completing the square, find values of a and b such that:

i) $x^2 + 2x + 5 = (x + a)^2 + b$ **2**

ii) Hence sketch $y = x^2 + 2x + 5$ **2**

b) Consider the equation $x^2 + (k + 3)x + 9 = 0$.

For what values of k does the equation have:

i) equal roots? **2**

ii) distinct real roots? **1**

c) Find the value of a if $\sin 35^\circ = \cos(a + 30)^\circ$ **1**

d) Find the exact value of $\sin 120^\circ \cos 45^\circ$ **2**

e) Given that $\cos \theta = -\frac{4}{5}$ and $\sin \theta > 0$, find the exact value of $\tan \theta$ **2**

End of Question 7

Question 8 (12 Marks)

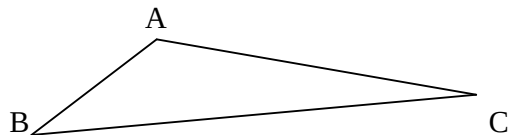
Use a Separate Booklet

Marks

a) Solve $|2x - 3| = 5x$

3

b)

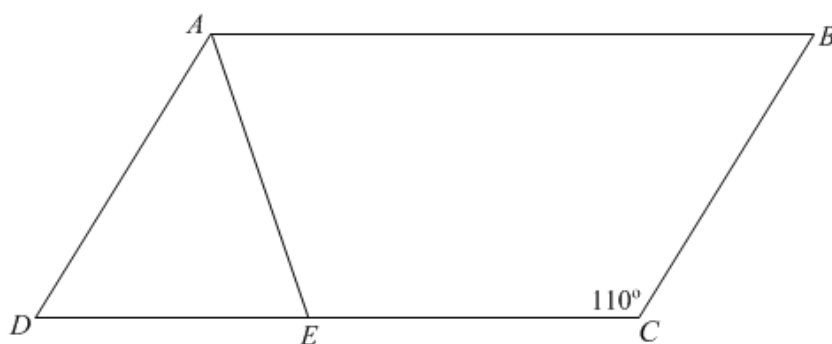
In the diagram, $AB = 12$ cm, $AC = 16$ cm and $BC = 21$ cm.i) Find, correct to the nearest minute, $\angle BAC$.**3**ii) Hence, find the area of $\triangle ABC$.**2**c) For the function $f(x) = \sqrt{4 - x^2}$, write down the natural domain and range.**2**d) Find the value of m in $x^2 + 2mx - 6 = 0$ if one of the roots is 2.**2****End of question 8**

Question 9 (12 Marks)

Use a Separate Booklet

Marks

- a) Find values of A, B and C if $x^2 + x - 2 \equiv A(x - 2)^2 + Bx + C$ 3
- b) $ABCD$ is a parallelogram and $AD = AE$. 3



Find $\angle DAE$, giving reasons.

- c) A golf ball is projected upwards so that its height (h) in metres, above the ground, is given by the formula $h = 90t - 15t^2$, where t is the time in seconds.
- i) When was the ball 75 metres above the ground? 2
- ii) How long before the ball falls to the ground? 2
- iii) What was the highest point reached by the ball? 2

End of Question 9

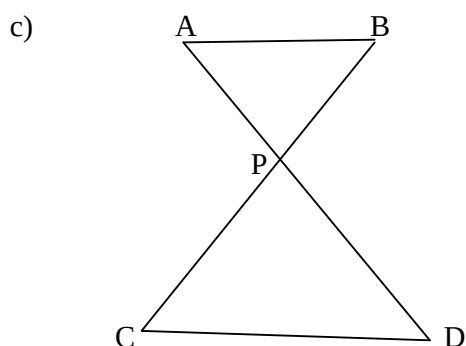
Question 10 (12 Marks)

Use a Separate Booklet

Marks

a) Prove that $\tan \theta + \cot \theta = \sec \theta \operatorname{cosec} \theta$ **3**

b) By completing the squares determine the centre and radius of the following circle: $x^2 - 2x + y^2 - 4y - 4 = 0$ **3**



In the diagram, $AB \parallel CD$, and AD and BC intersect at P.

Copy or trace the diagram onto your own paper.

If $\triangle ABP$ is similar to $\triangle DCP$ and $AB = 5$ cm, $CD = 7$ cm and $BC = 12$ cm, find the length of BP. **2**

d) Sketch $y = \frac{-1}{x+2} + 2$ **2**

e) Show that $2(2^k - 1) + 2^{k+1} = 2(2^{k+1} - 1)$ **2**

End of Exam

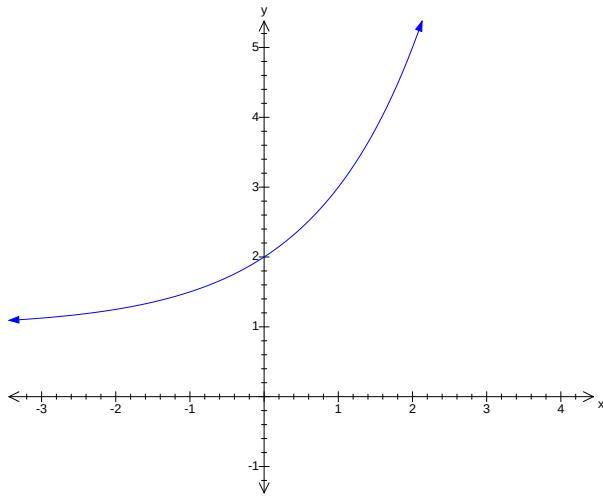
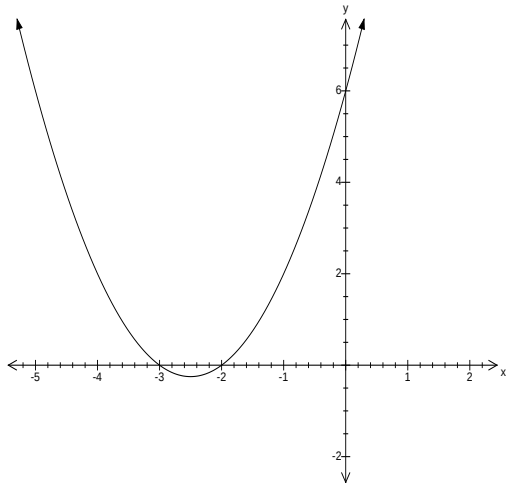


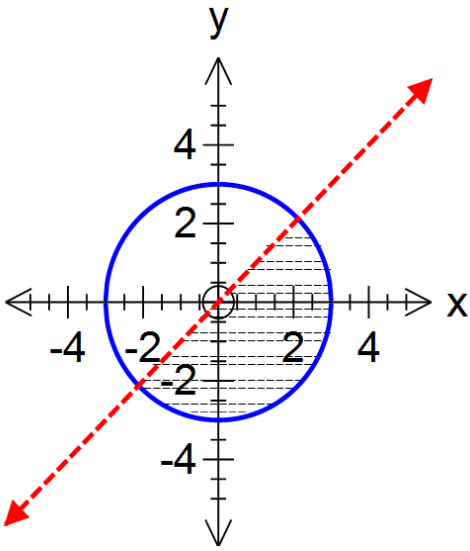
**Preliminary Final
EXAMINATION 2010**

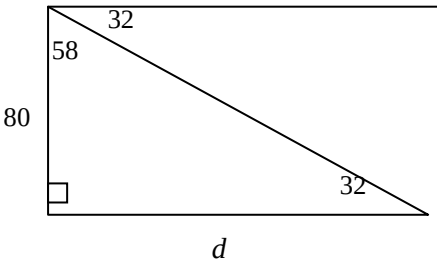
Mathematics

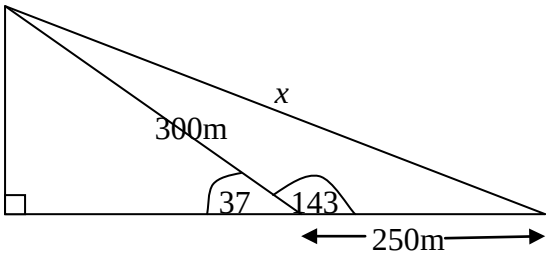
SOLUTIONS

Question 1		Preliminary HSC Examination- Mathematics	2010	
Part	Solution	Marks	Comment	
(a)	$(x-3)(2x+1)$ $= 2x^2 + x - 6x - 3$ $= 2x^2 - 5x - 3$	2		
(b)	$\sqrt{27} + \sqrt{12}$ $= 3\sqrt{3} + 2\sqrt{3}$ $= 5\sqrt{3}$ $= \sqrt{75}$ $\therefore A = 75$	2	1 for simplifying 1 for solution	
(c)	$x = 0.212121.....$ $100x = 21.21212121....$ $99x = 21$ $x = \frac{21}{99}$ $0.\dot{2}\dot{1} = \frac{7}{33}$	2		
(d)	$4x^2 - 16$ $= 4(x^2 - 4)$ $= 4(x+2)(x-2)$	2		
(e)	$\frac{\sqrt{3}}{3-\sqrt{2}} \times \frac{3+\sqrt{2}}{3+\sqrt{2}}$ $= \frac{3\sqrt{3}+\sqrt{6}}{9-2}$ $= \frac{3\sqrt{3}+\sqrt{6}}{7}$	2		
(f)	$\frac{2x-1}{5} = \frac{3x+2}{4}$ $4(2x-1) = 5(3x+2)$ $8x-4 = 15x+10$ $7x = -14$ $x = -2$	2		

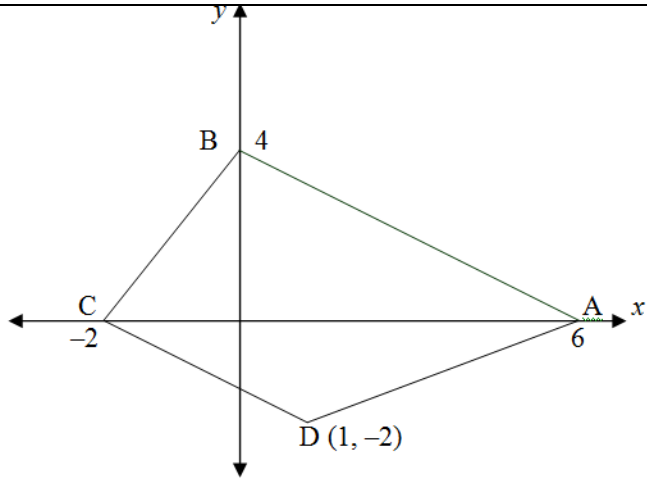
Question 2		Preliminary HSC Examination- Mathematics	2010	
Part	Solution	Marks	Comment	
(a)	$f(x) = 8x^3 - 7x$ $f(-x) = 8(-x)^3 - 7(-x)$ $= -8x^3 + 7x$ $= -(8x^3 - 7x)$ $= -f(x)$ As $f(-x) = -f(x)$, the function is odd.	2	2 for showing substitution for $-x$ and correct result of odd 1 for correct sub of $-x$	
(b) i)		2	2 correct shape, y intercept, another point (eg (1, 3)) and asymptote $y = 1$ 1 correct asymptote $y = 1$ and general shape	
(ii)	Domain: all real x Range: $y > 1$	2	1 mark for each	
(c)		3 2 1	Intercepts, vertex and scale correct Incorrect scale for the 6 and $-\frac{1}{4}$ Intercepts and shape	

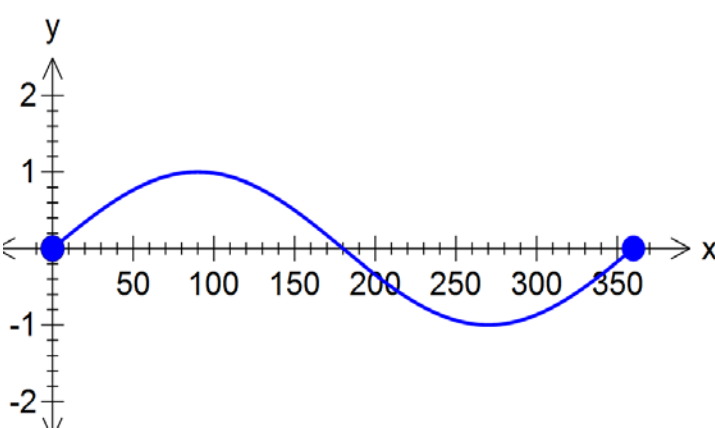
Question 2		Preliminary HSC Examination- Mathematics	2010	
Part	Solution	Marks	Comment	
(d)		3 2 1	for correct graphs incl dotted line for $y = x$ and correct region shaded incorrect region or $y = x$ not dotted correct circle	

Question 3		Preliminary HSC Examination- Mathematics	2010	
Part	Solution	Marks	Comment	
(a)	i)  ii) $\tan 58 = \frac{d}{80}$ $80 \tan 58 = d$ $d = 128.026...$ $= 128 \text{ m (nearest m)}$	1 2	1 for correct diagram clearly showing angle of depression 2 correct calculator answer 1 correct initial trig statement	

Question 3		Preliminary HSC Examination-Mathematics	2010	
Part	Solution	Marks	Comment	
(b)	i)  $x^2 = 300^2 + 250^2 - 2(300)(250)\cos(143)$ $x^2 = 272295.326\dots$ $x = 521.8192\dots$ $= 522 \text{ (nearest m)}$	2	2 for substituting correct data into the correct cosine rule and correct calculator answer 1 for substituting correct data into the correct cosine rule	
	ii) $\frac{\sin B}{300} = \frac{\sin 143}{522}$ $\sin B = 0.34587\dots$ $B = 20.23\dots$ $= 20^\circ 14'$	2	2 for substituting correct data into the correct sine rule and correct calculator answer. Must use 522m as stated in the previous part (i) 1 for substituting correct data into the correct sine rule	
(c)	$2\cos^2\theta = 1$ $\cos^2\theta = \frac{1}{2}$ $\cos\theta = \pm\frac{1}{\sqrt{2}}$ $\therefore \theta = 45^\circ, 135^\circ, 225^\circ, 315^\circ$	3	3 for all four correct angles 2 for correctly answering 45° and 315° 1 for getting $\cos\theta = \frac{1}{\sqrt{2}}$	
(d)	$4^{2x-1} = 8$ $(2^2)^{2x-1} = 2^3$ $2^{4x-2} = 2^3$ $\therefore 4x - 2 = 3$ $x = \frac{5}{4}$	2	2 correct answer and setting out 1 for writing $2^{4x-2} = 2^3$	

Question 4		Preliminary HSC Examination- Mathematics	2010	
Part	Solution		Marks	Comment

Question 4		Preliminary HSC Examination- Mathematics	2010	
Part	Solution	Marks	Comment	
(a)	 (i) $m_{AB} = \frac{4}{2}$ $= 2$ $m = \tan \theta$ $\tan \theta = 2$ $\theta = \tan^{-1} 2$ $= 63.4349 \dots \dots^\circ$ $= 63^\circ \text{ (to nearest degree)}$ (ii) Midpoint of AB: $M\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$ $M\left(\frac{6+0}{2}, \frac{0+4}{2}\right)$ $M(3, 2)$ (iii) $m_{CD} = \frac{y_2 - y_1}{x_2 - x_1}$ $= \frac{-2 - 0}{1 - -2}$ $= \frac{-2}{3}$ Equation of line CD $y - y_1 = m(x - x_1)$ $y - 0 = \frac{-2}{3}(x + 2)$ $3y = -2x - 4$ $\therefore 2x + 3y + 4 = 0$	2	2 marks: correct solution 1 mark:: correct gradient, incorrect angle OR Incorrect gradient, Correct angle from mistake	
		1	1 mark: correct answer	
		2	2 marks: correct solution with correct working 1 mark: correct gradient, incorrect equation or not enough working.	

Question 4		Preliminary HSC Examination- Mathematics	2010	
Part	Solution	Marks	Comment	
	(iv) $d = \frac{ ax_1 + by_1 + c }{\sqrt{a^2 + b^2}}$ $= \frac{ 2(3) + 3(2) + 4 }{\sqrt{2^2 + 3^2}}$ $= \frac{16}{\sqrt{13}} \times \frac{\sqrt{13}}{\sqrt{13}}$ $= \frac{16\sqrt{13}}{13} \text{ units}$	2	2 marks: correct answer with correct working 1 mark: not rationalising denominator OR Not showing how final answer was found	
	(v) $AB^2 = 4^2 + 6^2$ $= 16 + 36$ $AB = \sqrt{52}$ $= 2\sqrt{13}$ $\text{Area} = \frac{1}{2}(a + b)h$ $= \frac{1}{2}(\sqrt{13} + 2\sqrt{13}) \times \frac{16\sqrt{13}}{13}$ $= \frac{1}{2} \times 3\sqrt{13} \times \frac{16\sqrt{13}}{13}$ $= 24 \text{ units}^2$	3	3 marks: correct answer with correct working 2 marks: distance correct incorrect area formula correct area from mistake 1 mark: correct distance	
	(b) 	2		

Question 5		Preliminary HSC Examination- Mathematics	2010	
Part	Solution	Marks	Comment	
(a)	$\angle ACB = 60^\circ$ (vert.opp. $\angle DCE$) $\angle ABC = 62^\circ$ (angle sum of triangle = 180°) $\theta = 62^\circ$ (alternate angle to $\angle ABC$) $\phi = 58^\circ$ (alternate angle to $\angle BAC$)	2		
(b)	$f(-1) = 2(-1) - 3$ $= -2 - 3$ $= -5$ $f(1) = -3$ $f(-1) + f(1) = -5 + -3$ $= -8$	2		
(c)	$a^3 + 8$ $= (a + 2)(a^2 - 2a + 4)$	2		
(d)	i) $S = 180n - 360$ $3960 = 180n - 360$ $4320 = 180n$ $24 = n$ ii) $3960 \div 24 = 165^\circ$	1 1		
(e)	$\angle BAO = \angle OCD$ ($= 90^\circ$ given) $\angle BOA = \angle COD$ (vertically opposite $\angle =$) $BO = OD$ (given) $\therefore \triangle AOB \equiv \triangle COD$ (AAS)	2		
(f)	$ 3x - 5 \leq 2$ $-2 \leq 3x - 5 \leq 2$ $3 \leq 3x \leq 7$ $1 \leq x \leq \frac{7}{3}$	2		

Question 6		Preliminary HSC Examination- Mathematics	2010
Part	Solution	Marks	Comment
(a)	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ $= \frac{-(-6) \pm \sqrt{(-6)^2 - 4 \times 1 \times 7}}{2 \times 1}$ $= \frac{6 \pm \sqrt{36 - 28}}{2}$ $= \frac{6 \pm \sqrt{8}}{2}$ $= \frac{6 \pm 2\sqrt{2}}{2}$ $= 3 \pm \sqrt{2}$ $\approx 1.59, 4.41$	3	
(b)	$(5^2)^x - 26(5^x) + 25 = 0$ Let $m = 5^x$ $m^2 - 26m + 25 = 0$ $(m - 25)(m - 1) = 0$ $m = 25, m = 1$ $5^x = 25, 5^x = 1$ $\therefore x = 2, x = 0$	3	
(c)	Need to show that $\Delta < 0$ and $a < 0$ (negative definite) $b^2 - 4ac = 1^2 - (4 \times -3 \times -1)$ $= 1 - (12)$ $= -11$ $a = -3 < 0$ $\therefore$ Expression is negative definite and will always be less than zero for all x values.	2	

Question 6		Preliminary HSC Examination- Mathematics	2010	
Part	Solution	Marks	Comment	
(d)	i) $\alpha + \beta = -\frac{b}{a}$ $= -\frac{5}{2}$	1		
	ii) $\alpha\beta = \frac{c}{a}$ $= -\frac{3}{2}$	1		
	iii) $\frac{1}{a} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta}$ $= -\frac{5}{2} \div -\frac{3}{2}$ $= -\frac{5}{2} \times -\frac{2}{3}$ $= \frac{5}{3}$	2		

Question 7		Preliminary HSC Examination- Mathematics	2010	
Part	Solution	Marks	Comment	
(a)	i) $x^2 + 2x + 5 = (x^2 + 2x + 1) + 5 - 1$ $= (x+1)^2 + 4$ $\therefore a = 1, b = 4$	2		
	ii)	2		

Question 7		Preliminary HSC Examination- Mathematics		2010	
Part	Solution		Marks	Comment	
(b)	i) $x^2 + (k+3)x + 9 = 0$ $b^2 - 4ac = 0$ $(k+3)^2 - 4 \times 1 \times 9 = 0$ $k = 3, -9$ $k^2 + 6k + 9 - 36 = 0$ $k^2 + 6k - 27 = 0$ $(k+9)(k-3) = 0$ $k = 3, -9$		2		
			1		
	ii) $b^2 - 4ac > 0$ $(k+9)(k-3) > 0$ $k < -9, k > 3$		1		
	c) $\cos(90 - 35) = \cos(a + 30)$ $\therefore 90 - 35 = a + 30$ $a = 25$		1		
d)	$\sin 120^\circ \cos 45^\circ$ $= \frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{2}}$ $= \frac{\sqrt{3}}{2\sqrt{2}}$		2		
(e)	If cos is negative and sin is positive, angle is in second quadrant and tan is negative. $\tan \theta = -\frac{3}{4}$		2		

Question 8

a)	$ 2x-3 =5x$ $2x-3=5x \quad \text{or} \quad -(2x-3)=5x$ $3x=-3 \quad \text{or} \quad -2x+3=5x$ $x=-1 \quad \text{or} \quad 7x=3$ $= \frac{3}{7}$ Testing $x = -1$ $LHS = 2x-3 = 2(-1)-3 $ $= -5 $ $= 5$ $RHS = 5(-1)$ $= -5$ $\frac{3}{7}$ Testing $x = \frac{3}{7}$ $LHS = 2x-3 = \left 2\left(\frac{3}{7}\right) - 3 \right $ $= \left -2\frac{1}{7} \right $ $= 2\frac{1}{7}$ $RHS = 5 \times \left(\frac{3}{7}\right)$ $= 2\frac{1}{7}$ $\therefore x = \frac{3}{7}$ is a solution	3	
b)	i) $\cos A = \frac{12^2 + 16^2 - 21^2}{2 \times 12 \times 16}$ $= -0.10677....$ $A = 96^{\circ}8'$ ii) $\text{Area} = \frac{1}{2} \times 12 \times 16 \times \sin 96^{\circ}8'$ $= 95.451...$ $= 95.5 \text{ cm}^2 (1dp)$	3 2	

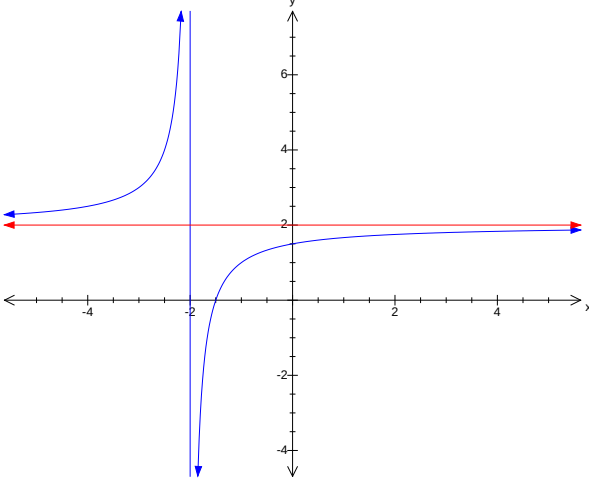
c)	Domain: $-2 \leq x \leq 2$ Range: $0 \leq y \leq 2$	2	
d)	If 2 is a root then it satisfies the equation: $2^2 + 2 \times m \times 2 - 6 = 0$ $4 + 4m - 6 = 0$ $4m - 2 = 0$ $m = \frac{2}{4}$ $= \frac{1}{2}$	2	

Question 9		Preliminary HSC Examination- Mathematics		2010	
Part	Solution	Marks	Comment		
(a)	$LHS = x^2 + x - 2$ $RHS = A(x - 2)^2 + Bx + C$ $= Ax^2 - 4Ax + 4A + Bx + C$ $= Ax^2 - x(4A - B) + 4A + C$ Equating coefficients, $A = 1$ $-(4A - b) = 1$ $4A + C = -2$ $-4 + B = 1$ $4 + C = -2$ $B = 5$ $C = -6$	3	3 for correct solutions 2 for two correct solutions 1 for one correct solution		
(b)	$\angle ADE + \angle BCD = 180^\circ$ (cointerior angles $AD \parallel AE$) $\angle ADE = 180^\circ - 170^\circ$ $= 70^\circ$ $\angle AED = \angle ADE$ (opposite equal sides of isosceles $\triangle AED$) $\angle AED = 70^\circ$ $\therefore \angle DAE = 180^\circ - \angle ADE - \angle AED$ (angle sum of a triangle) $= 180^\circ - 70^\circ - 70^\circ$ $= 40^\circ$	3	3 for showing clear logical working with correct reasoning and correct answer 2 one incorrect step or reason 1 only one step correctly completed		

Question 9		Preliminary HSC Examination- Mathematics	2010	
Part	Solution	Marks	Comment	
(c)	i) $\text{let } h = 75,$ $75 = 90t - 15t^2$ $15t^2 - 90t + 75 = 0$ $t^2 - 6t + 5 = 0$ $(t - 5)(t - 1) = 0$ $\therefore$ the ball is 75m at 1 second and 5 seconds	2	2 for correct answer with working 1 for creating the correct quadratic and putting it equal to zero ready to solve	
	ii) $\text{let } h = 0,$ $0 = 90t - 15t^2$ $0 = 15t(6 - t)$ $t = 0, 6$ $\therefore$ the ball falls to the ground after 6 seconds	2	2 for correct answer with working 1 for creating the correct factorised quadratic and putting it equal to zero ready to solve	
	(iii) $\text{let } t = 3,$ $h = 90(3) - 15(3)^2$ $= 270 - 135$ $= 135$ $\therefore$ the highest point reached by the ball is 135m.		2 for correct answer with working 1 for knowing to substitute three seconds to find highest point	

Question 9		Preliminary HSC Examination- Mathematics	2010	
Part	Solution	Marks	Comment	
(a)	$LHS = x^2 + x - 2 \quad RHS = A(x - 2)^2 + Bx + C$ $= Ax^2 - 4Ax + 4A + Bx + C$ $= Ax^2 - x(4A - B) + 4A + C$ Equating coefficients, $A = 1 \quad -(4A - B) = 1 \quad 4A + C = -2$ $\quad \quad -4 + B = 1 \quad 4 + C = -2$ $\quad \quad \quad B = 5 \quad \quad C = -6$	3	3 for correct solutions 2 for two correct solutions 1 for one correct solution	

Question 9		Preliminary HSC Examination- Mathematics	2010	
Part	Solution	Marks	Comment	
(b)	$\angle ADE + \angle BCD = 180^\circ$ (cointerior angles $AD \parallel AE$) $\angle ADE = 180^\circ - 170^\circ$ $= 70^\circ$ $\angle AED = \angle ADE$ (opposite equal sides of isosceles $\triangle AED$) $\angle AED = 70^\circ$ $\therefore \angle DAE = 180^\circ - \angle ADE - \angle AED$ (angle sum of a triangle) $= 180^\circ - 70^\circ - 70^\circ$ $= 40^\circ$	3	3 for showing clear logical working with correct reasoning and correct answer 2 one incorrect step or reason 1 only one step correctly completed	
(c)	i) let $h = 75$, $75 = 90t - 15t^2$ $15t^2 - 90t + 75 = 0$ $t^2 - 6t + 5 = 0$ $(t - 5)(t - 1) = 0$ $\therefore$ the ball is 75m at 1 second and 5 seconds	2	2 for correct answer with working 1 for creating the correct quadratic and putting it equal to zero ready to solve	
	ii) let $h = 0$, $0 = 90t - 15t^2$ $0 = 15t(6 - t)$ $t = 0, 6$ $\therefore$ the ball falls to the ground after 6 seconds	2	2 for correct answer with working 1 for creating the correct factorised quadratic and putting it equal to zero ready to solve	
	(iii) let $t = 3$, $h = 90(3) - 15(3)^2$ $= 270 - 135$ $= 135$ $\therefore$ the highest point reached by the ball is 135m.		2 for correct answer with working 1 for knowing to substitute three seconds to find highest point	

a)	Question 10 Solutions $LHS = \frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta}$ $= \frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta \cos \theta}$ $= \frac{1}{\sin \theta \cos \theta}$ $= \sec \theta \cos ec \theta$ $= RHS$	Marks 3 2 1	Comment Correct solution first 2 steps first step only
b)	$x^2 - 2x + 1 + y^2 - 4y + 4 = 4 + 1 + 4$ $(x-1)^2 + (y-2)^2 = 9$ centre (1, 2) radius 3	3 2 1	Correct working, radius and centre 1 error correct centre and radius from poor attempt correct completing square for one letter
c)	$\frac{x}{12-x} = \frac{5}{7}$ (corresponding sides in similar triangles are in the same ratio) $7x = 60 - 5x$ $12x = 60$ $x \text{ or } BP = 5$	2 1	answer plus reason answer only
d)		2 1	both asymptotes, intercepts or point on arms and shape both asymptotes or 1 error
e)	$LHS = 2^{k+1} - 2 + 2^{k+1}$ $= 2(2^{k+1}) - 2$ $= 2(2^{k+1} - 1)$ $= RHS$	2 1	both of the first 2 lines of working first line of working only before giving result